

Cisco Networking

Command Reference Guide

A Comprehensive Guide to Router and Switch Configuration

Overview:

This document serves as a comprehensive reference guide for Cisco IOS commands used in configuring and managing Cisco routers and switches. It covers essential networking concepts and practical implementations including:

- **Basic Device Configuration** - Initial setup, IP addressing, and device management
- **Routing Protocols** - Static and dynamic routing using RIP
- **VLAN Configuration** - Virtual LAN setup and inter-VLAN routing
- **Security Implementation** - Access control lists and password protection
- **Network Address Translation** - NAT configuration with pools and overloading
- **DHCP Services** - Dynamic host configuration and relay agents

Designed for network administrators, students, and IT professionals working with Cisco networking equipment in enterprise and laboratory environments.

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1 Basic Commands

This section covers fundamental Cisco IOS commands necessary for device connectivity testing and initial interface configuration.

1.1 Network Connectivity Testing

The ping command is used to test IP connectivity between devices by sending ICMP echo requests.

```
ping <ip_address>
```

Purpose: Verifies network connectivity and measures round-trip time (RTT) for packets.

1.2 Interface IP Address Configuration

Configuring IP addresses on router interfaces is essential for network communication. Each interface must have an IP address and subnet mask assigned.

```
enable
configure terminal
interface fastethernet0/0
ip address 192.168.0.101 255.255.255.0
no shutdown
```

Explanation:

- `enable` - Enters privileged EXEC mode
- `configure terminal` - Enters global configuration mode
- `interface fastethernet0/0` - Selects the specific interface
- `ip address` - Assigns IP address and subnet mask
- `no shutdown` - Activates the interface (interfaces are shutdown by default)

1.3 Viewing the Routing Table

The routing table displays all known networks and the paths to reach them.

```
enable
show ip route
```

Note: If a network is not in the routing table, packets destined for that network will be dropped. Administrative distance indicates route trustworthiness (lower values are more trusted).

2 Routing Configuration

Routing enables communication between different network segments. Cisco routers support both static and dynamic routing protocols.

2.1 Static Routing

Static routes are manually configured and remain constant unless changed by an administrator.

```
enable
configure terminal
ip route 192.168.3.0 255.255.255.0 <exit_interface or next_hop_ip>
```

Use Case: Best for small networks or specific routing requirements where routes don't change frequently.

2.2 Dynamic Routing with RIP

Routing Information Protocol (RIP) automatically shares routing information between routers.

```
enable
configure terminal
router rip
version 2
network 192.168.0.0
network 192.168.1.0
no auto-summary
```

Explanation:

- `router rip` - Enables RIP routing protocol
- `version 2` - Uses RIPv2 (supports VLSM and classless routing)
- `network` - Specifies networks to advertise
- `no auto-summary` - Disables automatic route summarization

3 VLAN Configuration

Virtual LANs (VLANs) segment broadcast domains at Layer 2, improving network performance and security.

3.1 Viewing Existing VLANs

```
enable
show vlan
```

Note: VLAN 1 is the default VLAN and cannot be deleted.

3.2 Creating VLANs

```
enable
configure terminal
vlan 10
name faculty
```

Purpose: Creates logical network segments within a physical switch infrastructure.

3.3 VLAN Interface Assignment

3.3.1 Access Mode (Single VLAN)

Access ports connect end devices (PCs, printers) to a single VLAN.

```
enable
configure terminal
interface fastethernet1/1
switchport mode access
switchport access vlan 10
```

3.3.2 Trunk Mode (Multiple VLANs)

Trunk ports carry traffic for multiple VLANs between switches or to routers.

```
enable
configure terminal
interface fastethernet0/1
switchport mode trunk
switchport trunk native vlan 1
switchport trunk allowed vlan 1,10,30
```

Explanation:

- `switchport mode trunk` - Configures port as trunk
- `trunk native vlan` - Specifies untagged VLAN
- `trunk allowed vlan` - Restricts which VLANs can traverse the trunk

3.4 Inter-VLAN Routing (Router-on-a-Stick)

Inter-VLAN communication requires a Layer 3 device (router) since VLANs are isolated at Layer 2.

3.4.1 Router Configuration

```
enable
configure terminal
interface fastethernet0/0.10
encapsulation dot1q 10
ip address 192.168.10.1 255.255.255.0
exit
interface fastethernet0/0
no shutdown
```

Key Concepts:

- Subinterfaces (e.g., fa0/0.10) represent logical interfaces
- `encapsulation dot1q` enables IEEE 802.1Q VLAN tagging
- Each subinterface serves as the default gateway for its VLAN

3.4.2 Switch Configuration

```
enable
configure terminal
interface fastethernet0/1
switchport mode trunk
switchport trunk native vlan 1
switchport trunk allowed vlan 1,10,30
```

4 Security and Access Control

4.1 Password Configuration

4.1.1 Enable Password

Protects privileged EXEC mode access.

```
enable
configure terminal
enable secret <password>
```

Note: Use `enable secret` instead of `enable password` as it provides MD5 encryption.

4.1.2 Console Password

Secures physical console access.

```
enable
configure terminal
line console 0
password <password>
login
exit
```

4.2 Device Identification

```
enable
configure terminal
hostname Router1
```

4.3 Virtual Terminal (Telnet) Access

Enables remote management via Telnet.

```
enable
configure terminal
line vty 0 4
password <password>
login
```

Connecting from PC:

```
telnet <IP_Address>
```

Security Note: Telnet transmits data in cleartext. SSH is recommended for production environments.

5 Access Control Lists (ACLs)

ACLs filter network traffic based on defined criteria, providing security and traffic management.

5.1 Standard ACLs

Standard ACLs filter traffic based solely on source IP address (1-99, 1300-1999).

```
enable
configure terminal
access-list 1 deny host 192.168.1.1
access-list 1 permit host 192.168.1.2
do show access-list
```

Applying to Interface:

```
enable
configure terminal
interface fastethernet4/0
ip access-group 1 in
```

Important: ACLs have an implicit "deny all" at the end. Always include permit statements.

5.1.1 Deleting ACLs

```
enable
configure terminal
no access-list 1
```

5.2 Extended ACLs

Extended ACLs provide granular control (100-199, 2000-2699) filtering by:

- Source and destination IP addresses
- Protocol type (TCP, UDP, ICMP)
- Source and destination port numbers
- Packet types

```
enable
configure terminal
access-list 100 deny icmp 192.168.1.0 0.0.0.255 host 192.168.3.1 echo
access-list 100 permit icmp 192.168.1.0 0.0.0.255 host 192.168.3.2 echo
```

Applying to Interface:

```
enable
configure terminal
interface fastethernet4/0
ip access-group 100 in
```

Best Practice: Place extended ACLs close to the source to reduce unnecessary traffic on the network.

6 Network Address Translation (NAT)

NAT translates private IP addresses to public addresses, enabling internet connectivity while conserving public IP addresses.

6.1 Private IP Address Ranges

- **Class A:** 10.0.0.0 - 10.255.255.255
- **Class B:** 172.16.0.0 - 172.31.255.255
- **Class C:** 192.168.0.0 - 192.168.255.255

6.2 NAT with Interface Overload (PAT)

Port Address Translation (PAT) maps multiple private addresses to a single public IP using different ports.

```
enable
configure terminal
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface fastethernet4/0 overload
interface fastethernet0/0
ip nat inside
exit
interface fastethernet4/0
ip nat outside
exit
```

6.3 Viewing NAT Translations

```
enable
show ip nat translations
```

6.4 NAT with Address Pool

6.4.1 Creating NAT Pool

```
enable
configure terminal
ip nat pool pool_1 209.165.200.1 209.165.200.4 netmask 255.255.255.0
```

6.4.2 Configuring NAT with Pool

```
enable
configure terminal
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 pool pool_1 overload
interface fastethernet0/0
ip nat inside
exit
interface fastethernet4/0
ip nat outside
exit
```


6.4.3 Routing for NAT Pool (Other Router)

External routers need routes to the NAT pool addresses.

```
enable
configure terminal
ip route 209.165.200.0 255.255.255.0 fastethernet4/0
```

7 DHCP Configuration

Dynamic Host Configuration Protocol automatically assigns IP addresses and network parameters to devices.

7.1 Router as DHCP Server

```
enable
configure terminal
ip dhcp excluded-address 192.168.2.101
ip dhcp pool pool1_192.168.2.0
network 192.168.2.0 255.255.255.0
default-router 192.168.2.101
```

Explanation:

- **excluded-address** - Reserves IP addresses (typically for static assignments)
- **pool** - Creates a DHCP address pool
- **network** - Specifies network and subnet mask
- **default-router** - Sets the default gateway for clients

7.2 DHCP Relay Agent

DHCP relay forwards DHCP requests across network boundaries.

7.2.1 Relay Router Configuration

```
enable
configure terminal
interface fastethernet0/0
ip helper-address 192.168.3.101
```

7.2.2 Main DHCP Server Configuration

```
enable
configure terminal
ip dhcp excluded-address 192.168.4.101
ip dhcp pool pool2_192.168.4.0
network 192.168.4.0 255.255.255.0
default-router 192.168.4.101
exit
ip route 192.168.4.0 255.255.255.0 serial2/0
```

8 Advanced: RIP with NAT Pool Advertisement

Challenge: RIP only advertises networks present on active router interfaces. NAT pools don't automatically appear in RIP updates.

Solution: Create a loopback interface with an IP from the NAT pool range.

8.1 Configuration Steps

1. Configure RIP on all routers
2. On the NAT router:

```
enable
configure terminal

# Create ACL for NAT
access-list 1 permit 192.168.1.0 0.0.0.255

# Create NAT pool
ip nat pool pool_1 209.165.200.126 209.165.200.140 netmask 255.255.255.0

# Configure NAT
ip nat inside source list 1 pool pool_1 overload

# Create loopback interface
interface loopback 0
ip address 209.165.200.126 255.255.255.0
exit

# Advertise pool network via RIP
router rip
version 2
network 209.165.200.0
```

Benefit: The loopback interface makes the NAT pool network visible to RIP, allowing other routers to learn routes to the public IP addresses.